

ANDREW ZHENG

410-852-2128 | azheng15@umd.edu | [Personal Website](#) | [Github](#)

EDUCATION

University of Maryland | College Park

M.S. in Computer Science

Aug 2024 – May 2026

B.S. in Computer Science and Mathematics

Jan 2021 – May 2024

SKILL HIGHLIGHTS

Programming Languages: Python, C++, C, Java, SQL, Bash, LaTeX

Quantitative Skills: Linear Algebra, Numerical Optimization, PDEs, Statistical Modeling, Abstract Algebra, Calculus

Software & Tools: Git (Version Control), Pytorch, Pandas, Automated Testing (Unit/Integration/Regression)

Domains: Algorithm Design, Numerical Methods, High-Performance Computing, Model Validation, Machine Learning

WORK EXPERIENCE

Research Intern | University of Maryland, College Park

May 2024 – August 2024

Quantum Computing

- Designed and implemented efficient numerical algorithms in Python for computing simulation error bounds
- Validated implementation accuracy through reconciliation against known analytical results
- Maintained version-controlled codebase with documentation and change logs for reproducibility

Information Technology Intern | AARP

May 2023 – August 2023

Generative AI

- Led end-to-end development of a production chatbot system, translating business requirements into technical design
- Built scalable data collection and processing pipelines using Pandas, Selenium, and BeautifulSoup
- Trained and deployed large language models for production use using Transformers and Pytorch

Teaching Assistant | University of Maryland, College Park

Jan 2023 – May 2023

CMSC 250: Discrete Structures

- Communicated mathematical foundations (logic, proofs, combinatorics) to classes of 30+ students
- Created original instructional materials to complement lecture content: [\[Link\]](#)

RESEARCH AND PUBLICATIONS

Keypoint Reconstruction for Sports Analytics | *Computer Vision*

June 2026 – Current

- Designed and built an AI that provides feedback to improve ice hockey players

Guardian Models for LLMs | *Reinforcement Learning, verl*

Jan 2026 – Current

- Optimized large-scale model training for computational efficiency and scalability under hardware constraints
- Continuously improved model architecture and computational performance through iterative refactoring
- Advised under Professor Tom Goldstein, in collaboration with Monte Hoover

Self-Improvement for LLMs | *Reinforcement Learning, Monte-Carlo Methods*

Aug 2025 – Current

- Designed, built, and validated a quantitative framework for iterative test-time model improvement
- Developed automated testing and evaluation pipelines to ensure numerical correctness and reproducibility
- In collaboration with Dr. Zikui Cai
- Paper: [\[Link\]](#)

Quantum Simulation | *Algorithm Analysis, Numerical Methods*

Jan 2024 – Current

- Designed computational algorithms and a built software package for computing the error of theoretical quantum mechanics
- Analyzed accuracy-vs-computational-cost tradeoffs for higher-order numerical methods
- Advised by Professor Andrew Childs, in collaboration with Professor Jake Bringewatt, Dr. James Watson and David Kong

- Paper: [\[Link\]](#); code provided upon request

Noise Cancellation | *Pytorch, Deep Learning*

Jan 2023 - May 2023

- Built and optimized high-performance deep learning models for real-time signal processing on resource-constrained devices
- Paper and Code: [\[Paper\]](#) [\[Code\]](#)

SELECTED COURSEWORK

Computer Science: Advanced Numerical Optimization (CMSC764); Introduction to Compilers (CMSC430); Foundations of Deep Learning (CMSC720); Capstone in Machine Learning (CMSC673); Introduction to Machine Learning (CMSC422); Computer Vision (CMSC426); Advanced Topics in Quantum Computing (CMSC858Q); Introduction to Quantum Computing (CMSC457)

Mathematics: Advanced Linear Algebra (MATH405); Computational Methods (AMSC460); PDE's (MATH462); Complex Analysis (MATH463); Transform Methods (MATH464); Advanced Calculus II (MATH411); Abstract Algebra I (MATH600); Abstract Algebra (MATH403); Lie Algebra (MATH744); Number Theory (MATH406); Introduction to Topology (MATH432)

OTHER EXPERIENCES

Chess

- Achieved National Master (NM) Title
- Scholastic Maryland State Champion | 2014 - 2017, 2020
- Represented UMD Chess Team in Pan American Chess Championships | 2021, 2022

Ice Hockey

- Washington Little Caps AAA U18 | 2019 – 2020
- Howard County All-Star 2016-2020, MSHL all-state | 2018 – 2020